Answers microeconomics exam

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Question 1

a. Opportunity cost = the best forgone alternative = working: 8 × €14 = €112.

b. Tuition and the book are sunk costs: they have already been paid and won't come back, whatever she does on Saturday. So they should not be part of the trade-off.

Question 2

a. Cheaper batteries lower production costs, so supply shifts to the right. Along the demand curve there is a movement: at the lower price more is demanded.

b. Demand shifts to the right: the e-bike is a substitute for the car, and driving becomes more expensive.

Question 3

a. 200 − 4P = −40 + 2P → 6P = 240 → P = 40, Q = 40 (40,000 units).

b. Consumer surplus = ½ × 40 × 40 = 800 (× €1,000). Producer surplus = ½ × (40 − 20) × 40 = 400 (× €1,000).

Question 4

a. Qs = −40 + 2(P − 6) = −52 + 2P → 200 − 4P = −52 + 2P → P = 42, Q = 32.

b. Revenue = 6 × 32 = 192 (× €1,000). Deadweight loss = ½ × 6 × 8 = 24 (× €1,000).

c. Consumers pay €2 more, producers receive €4 less. Still, consumers bear most of it, because they pay the higher price at the till.

Question 5

a. E = dQ/dP × P/Q = −4 × 40/40 = −4.

b. |E| > 1, so elastic. If the price goes up, spending falls, because quantity falls proportionally more than the price rises.

Question 6

a. 12,000 / (3.50 − 1.50) = 6,000 loaves.

b. Profit = 8,000 × 2 − 12,000 = €4,000. ATC = (12,000 + 1.50 × 8,000) / 8,000 = 24,000 / 8,000 = €3.00.

Question 7

a. MR = 100 − 4Q = 20 → Q = 20, P = 60.

b. Profit = (60 − 20) × 20 = 800.

c. Under perfect competition P = MC = 20 → Q = 40. Deadweight loss = (60 − 20) × (40 − 20) = 800.

Question 8

a. Negative externality: marginal social cost lies above marginal private cost, so the market produces more than is socially optimal.

b. The government could introduce a permit system.

Question 9

a. Low is dominant for both (10 > 8 and 4 > 2). Nash equilibrium: (low, low).

b. Both high gives more profit (8 ; 8), but each firm has an incentive to deviate to low, so an agreement is not stable. That is a prisoner's dilemma.

Question 10

Market structure	Number of firms	Product	Barriers to entry	Pricing	Long-run economic profit
Perfect competition	Very many	Homogeneous	None	Price taker, P = MC	Zero
Monopolistic competition	Many	Differentiated	Low	Some market power, P > MC	Positive
Oligopoly	A few	Homogeneous or differentiated	High	Depends on rivals	Possibly positive
Monopoly	One	Unique	Very high	Price maker, P > MC	Possibly positive